

IIT HYDERABAD

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices and Circuits Laboratory Report

Experiment No: **04**



Course Name: Electronic Devices and Circuits Lab

Course Code: EE2301

Submitted by

Student 1 Name : Dhanush Sagar

Student 1 Roll No : EE25BTECH11021

Student 2 Name : Sreetej Darisy

Student 2 Roll No : EE25BTECH11018

Branch : Electrical Engineering

Semester : III

Submitted to

Gajendranath Chaudhury

September 5, 2026

Contents

1	Objective	3
2	Apparatus Required	3
3	Part A : DC Transfer Characteristic	3
3.1	Aim	3
3.2	Theory	3
3.3	Circuit	5
3.4	Observations	6
3.5	Plots	7
3.6	Results	8
4	Part B : Small-Signal Voltage Gain	8
4.1	Aim	8
4.2	Theory	8
4.3	Method 1	9
4.4	Method 2	10
4.4.1	Circuit Configuration	10
4.5	Observations	11
4.6	Relevant MOSFET Parameters from Experiment-3	12
4.7	Sample Calculations	12
4.7.1	Gain from AC Measurement	12
4.7.2	Maximum Gain	12
4.7.3	Method 1 Gain	13
4.8	Plots and Simulation	13
4.8.1	Below Cutoff ($V_{in} = 1V$)	13
4.8.2	Saturation ($V_{in} = 1.68V$)	14
4.8.3	Deep Triode ($V_{in} = 2V$)	14
5	Analysis	15
5.1	Bias Point for Maximum Gain	15
5.2	Gain in Cutoff and Deep Triode Regions	16
5.3	Comparison of Method 1 and Method 2	17
5.4	Effect of Increasing R_D	18
5.5	Relation Between the DC Transfer Slope and AC Gain	19
5.6	Comparison with the Illustrative Section 6 Values	19
5.7	Comparison of Theoretical Gain with Part A and Part B	20

6	Sources of Error	22
7	Conclusion	22

1 Objective

To construct an **NMOS** common-source amplifier with a resistive drain load, obtain its DC voltage transfer characteristics (VTC), evaluate the small-signal voltage gain at various DC operating bias points using two distinct analytical/experimental methods, and systematically compare the results

2 Apparatus Required

- NMOS transistor
- DC power supply
- Function generator
- DSO
- Digital multimeter (DMM)
- Drain resistor (R_D)
- Coupling capacitor
- Blocking resistor
- Potentiometer / programmable DC source
- Breadboard and connecting wires

3 Part A : DC Transfer Characteristic

3.1 Aim

To obtain the DC transfer characteristic (V_{OUT} vs V_{IN}) of an NMOS common-source amplifier with a resistive drain load and to identify the cutoff, saturation, and triode operating regions.

3.2 Theory

In an NMOS common-source amplifier, the source is grounded, the gate is driven by the input voltage V_{IN} , and the drain is connected to the supply voltage V_{DD} through a drain resistor R_D . The output voltage V_{OUT} is measured at the drain.

Since the source is grounded,

$$V_{GS} = V_{IN}$$

and the output voltage is given by

$$V_{OUT} = V_{DD} - I_D R_D.$$

The operating region of the NMOS depends on the applied input voltage V_{IN} .

Cutoff Region

When

$$V_{IN} < V_{TH},$$

the NMOS is OFF and

$$I_D \approx 0.$$

Therefore,

$$V_{OUT} \approx V_{DD}.$$

Thus, the output voltage remains nearly constant at the supply voltage in the cutoff region.

Saturation Region

When

$$V_{GS} > V_{TH}$$

and

$$V_{DS} \geq V_{GS} - V_{TH},$$

the NMOS operates in the saturation region. This is the amplifying region of the common-source amplifier.

As V_{IN} increases, the drain current I_D increases, resulting in a larger voltage drop across R_D . Hence,

$$V_{OUT} = V_{DD} - I_D R_D$$

decreases with increasing V_{IN} .

The slope of the DC transfer characteristic in this region is related to the voltage gain:

$$A_v = \frac{dV_{OUT}}{dV_{IN}}.$$

Triode Region

When

$$V_{DS} < V_{GS} - V_{TH},$$

the NMOS enters the triode region. In this region, the drain current increases significantly with increasing V_{IN} , causing the output voltage to become very small.

Therefore,

$$V_{OUT} \rightarrow 0.$$

3.3 Circuit

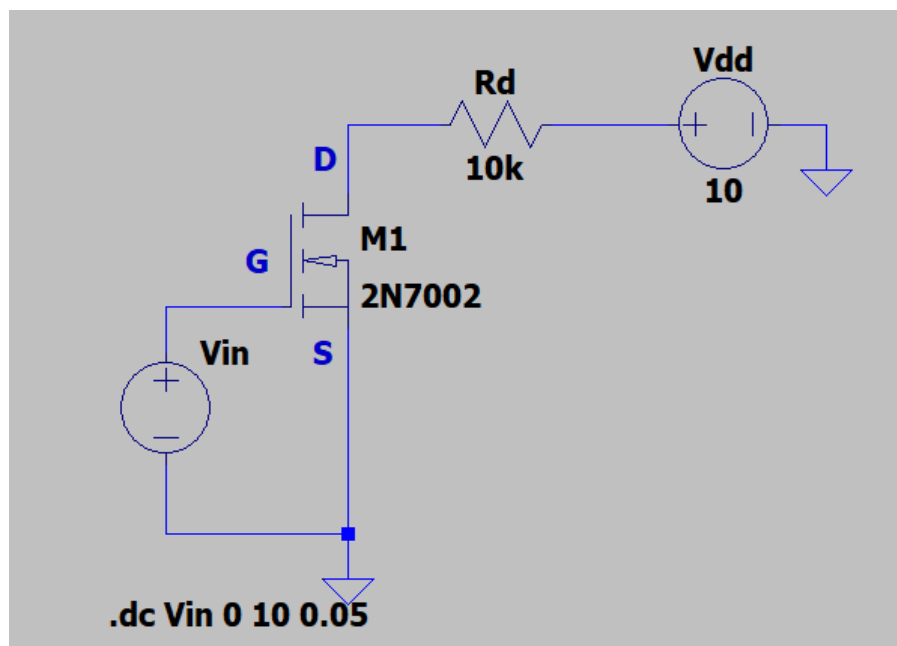


Figure 1: LTspice Schematic Circuit

3.4 Observations

Table 1: Experimental DC Transfer Characteristic Dataset

V_{IN} (V)	V_{OUT} (V)	Operating Region
0.00	9.9910	Cut-off ($V_{IN} < V_{TH}$)
0.20	9.9910	Cut-off
0.60	9.9910	Cut-off
0.80	9.9910	Cut-off
1.00	9.9900	Cut-off
1.20	9.9870	Cut-off
1.40	9.8970	Near Threshold ($V_{IN} \approx V_{TH}$)
1.45	9.7880	Saturation
1.50	9.5600	Saturation
1.55	9.0930	Saturation
1.60	8.2000	Saturation
1.65	6.5800	Saturation
1.70	3.8100	Saturation
1.75	0.1740	Triode ($V_{DS} < V_{GS} - V_{TH}$)
1.80	0.0572	Triode
1.85	0.0378	Deep Triode
1.90	0.0243	Deep Triode
1.95	0.0182	Deep Triode
2.00	0.0144	Deep Triode
2.05	0.0117	Deep Triode
2.10	0.0098	Deep Triode
2.20	0.0077	Deep Triode
2.25	0.0069	Deep Triode
2.50	0.0047	Deep Triode
3.00	0.0034	Deep Triode
5.00	0.0025	Deep Triode
7.50	0.0022	Deep Triode
10.00	0.0021	Deep Triode

3.5 Plots

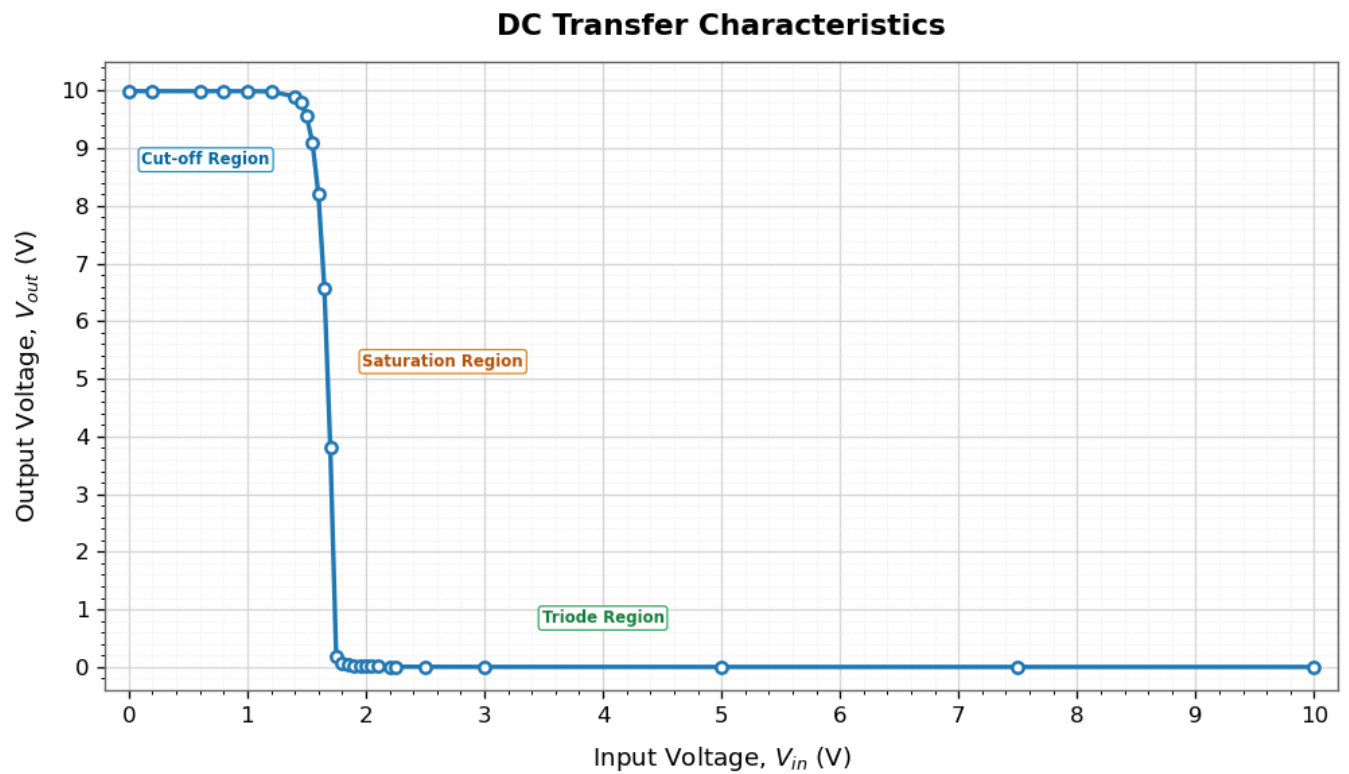


Figure 2: Experimental Plot

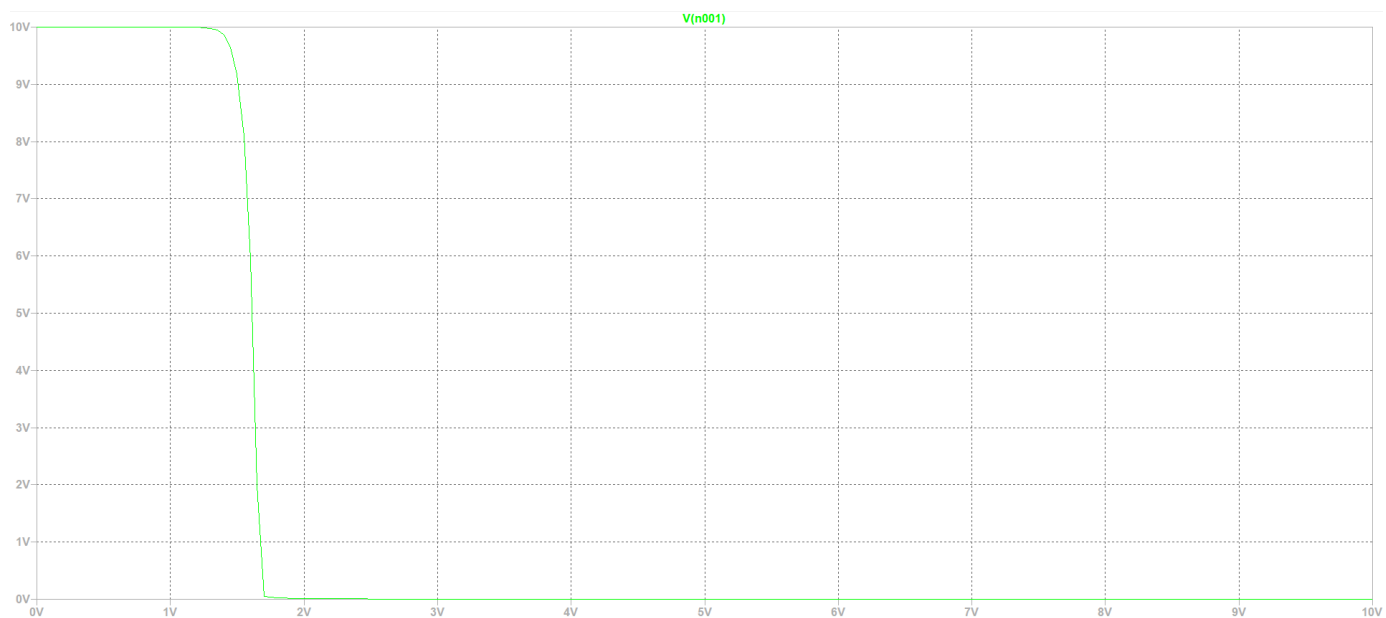


Figure 3: Simulation Plot

3.6 Results

The DC transfer characteristic (V_{OUT} vs V_{IN}) of the NMOS common-source amplifier was obtained experimentally. The characteristic shows three distinct operating regions: cutoff, saturation (amplifying), and triode.

For the measured circuit,

$$V_{DD} = 10 \text{ V}, \quad R_D = 10 \text{ k}\Omega.$$

The following regions were observed:

- **Cutoff region:** $V_{OUT} \approx 9.99 \text{ V}$ for low values of V_{IN} .
- **Saturation region:** V_{OUT} decreases rapidly as V_{IN} increases, approximately from $V_{IN} = 1.45 \text{ V}$ to 1.70 V .
- **Triode region:** For $V_{IN} \gtrsim 1.75 \text{ V}$, V_{OUT} becomes very small and approaches 0 V .

Thus, the experimental V_{OUT} - V_{IN} characteristic clearly demonstrates the transition of the NMOS through the cutoff, saturation, and triode regions.

4 Part B : Small-Signal Voltage Gain

4.1 Aim

To determine the small-signal voltage gain of the NMOS common-source amplifier at different DC bias points using the numerical slope of the measured DC transfer characteristic

4.2 Theory

For a common-source amplifier, the small-signal voltage gain is

$$A_v = -\frac{v_{out}}{v_{in}}. \quad (1)$$

The negative sign indicates the 180° phase inversion between input and output.

For a MOSFET operating in saturation, the approximate voltage gain is

$$A_v \approx -g_m(R_D \parallel r_d), \quad (2)$$

where

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \quad (3)$$

is the transconductance and r_d is the small-signal drain resistance.

The saturation condition is approximately

$$V_{DS} \geq V_{GS} - V_T. \quad (4)$$

When

$$V_{DS} < V_{GS} - V_T,$$

the MOSFET enters the triode region.

The DC transfer characteristic is

$$V_{OUT} = f(V_{IN}). \quad (5)$$

For a small AC signal around a bias point Q ,

$$v_{out} = \left. \frac{dV_{OUT}}{dV_{IN}} \right|_Q v_{in}. \quad (6)$$

Therefore,

$$\boxed{A_v = \frac{v_{out}}{v_{in}} = \left. \frac{dV_{OUT}}{dV_{IN}} \right|_Q}. \quad (7)$$

Thus, the local slope of the DC transfer curve represents the small-signal AC gain at the same operating point.

4.3 Method 1

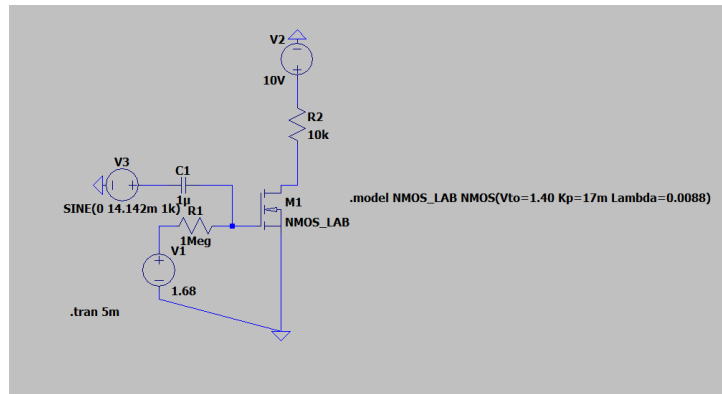
The gain is evaluated at selected DC bias points from the measured V_{OUT} versus V_{IN} characteristic.

Table 2: Calculation of Small-Signal Voltage Gain

$V_{IN,1}$ (V)	$V_{IN,2}$ (V)	A_v
0.00	0.20	0.00000
0.20	0.60	0.00000
0.60	0.80	0.00000
0.80	1.00	-0.00500
1.00	1.20	-0.01500
1.20	1.40	-0.45000
1.40	1.45	-2.18000
1.45	1.50	-4.56000
1.50	1.55	-9.34000
1.55	1.60	-17.86000
1.60	1.65	-32.40000
1.65	1.70	-55.40000
1.70	1.75	-72.72000
1.75	1.80	-2.33600
1.80	1.85	-0.38800
1.85	1.90	-0.27000
1.90	1.95	-0.12200
1.95	2.00	-0.07600
2.00	2.05	-0.05400
2.05	2.10	-0.03800
2.10	2.20	-0.02100
2.20	2.25	-0.01600
2.25	2.50	-0.00900
2.50	3.00	-0.00300
3.00	5.00	-0.00045
5.00	7.50	-0.00012
7.50	10.00	-0.00004

4.4 Method 2

4.4.1 Circuit Configuration

**Figure 4:** LTspice Schematic Circuit

4.5 Observations

Table 3: Measured AC output voltage and calculated gain

$V_{IN,DC}$ (V)	$V_{OUT,RMS}$ (mV)	$V_{OUT,amp}$ (mV)	A_v (V/V)
1.30	6.69	9.46	-0.669
1.35	8.30	11.74	-0.830
1.38	15.40	21.78	-1.540
1.40	19.40	27.44	-1.940
1.42	24.00	33.94	-2.400
1.44	30.50	43.13	-3.050
1.46	39.80	56.29	-3.980
1.48	52.20	73.82	-5.220
1.50	68.50	96.87	-6.850
1.52	89.90	127.14	-8.990
1.54	117.10	165.60	-11.710
1.56	151.00	213.55	-15.100
1.58	194.00	274.36	-19.400
1.60	247.00	349.31	-24.700
1.62	312.80	442.12	-31.280
1.64	391.00	552.96	-39.100
1.66	484.00	684.48	-48.400
1.68	593.00	838.63	-59.300
1.70	715.10	1011.30	-71.510
1.72	852.00	1204.31	-85.200
1.74	544.00	769.33	-54.400
1.76	39.50	55.86	-3.950
1.78	12.60	17.82	-1.260
1.80	6.93	9.80	-0.693
1.82	4.55	6.44	-0.455
1.84	3.247	4.59	-0.3247
1.86	2.93	4.14	-0.293
1.88	1.92	2.72	-0.192
1.90	1.55	2.19	-0.155

4.6 Relevant MOSFET Parameters from Experiment-3

The following measured parameters from the previous MOSFET experiment are used for comparison:

Table 4: MOSFET parameters used in the analysis

Parameter	Measured / Calculated Value
Threshold voltage, onset method	1.40 V
Threshold voltage, $\sqrt{I_D}$ extrapolation	1.48 V
Transconductance, g_m at $V_{GS} = 2.0$ V	57.0 mS
Drain resistance, r_d at $V_{GS} = 2.0$ V	13.33 k Ω
Drain resistance, r_d at $V_{GS} = 1.8$ V	181.82 k Ω
Drain resistance, r_d at $V_{GS} = 1.6$ V	1.43 M Ω
Amplification factor, μ	760

The measured transfer characteristic also gives

$$I_D(1.7 \text{ V}) = 0.660 \text{ mA}$$

and

$$I_D(1.8 \text{ V}) = 1.844 \text{ mA}.$$

4.7 Sample Calculations

4.7.1 Gain from AC Measurement

At

$$V_{IN,DC} = 1.60 \text{ V},$$

the measured output voltage is

$$V_{OUT,RMS} = 247 \text{ mV}.$$

Therefore,

$$A_v = -\frac{247}{10}$$

$$\boxed{A_v = -24.70}.$$

4.7.2 Maximum Gain

At

$$V_{IN,DC} = 1.72 \text{ V},$$

$$V_{OUT,RMS} = 852 \text{ mV}.$$

Hence,

$$A_v = -\frac{852}{10}$$

$$\boxed{A_v = -85.20}.$$

Thus,

$$\boxed{|A_v|_{\max} = 85.20}.$$

4.7.3 Method 1 Gain

For the interval 1.70–1.75 V,

$$A_v = \frac{V_{OUT}(1.75) - V_{OUT}(1.70)}{1.75 - 1.70}.$$

Using the measured DC output values,

$$A_v = \frac{0.174 - 3.810}{0.05}$$

$$\boxed{A_v = -72.72}.$$

4.8 Plots and Simulation

4.8.1 Below Cutoff ($V_{in} = 1V$)

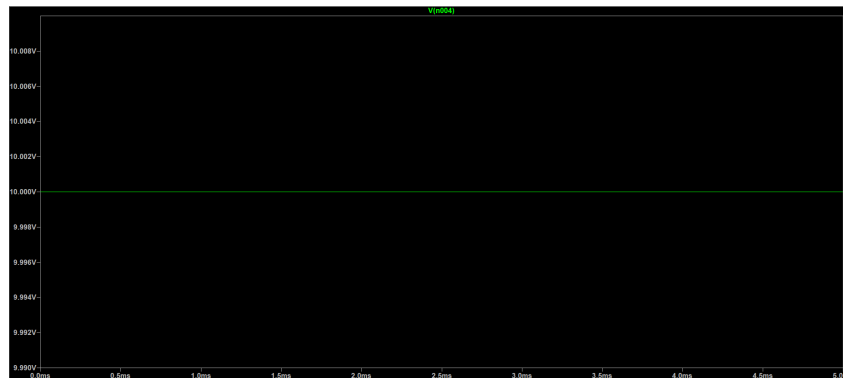


Figure 5: Simulation Plot

4.8.2 Saturation ($V_{in} = 1.68V$)

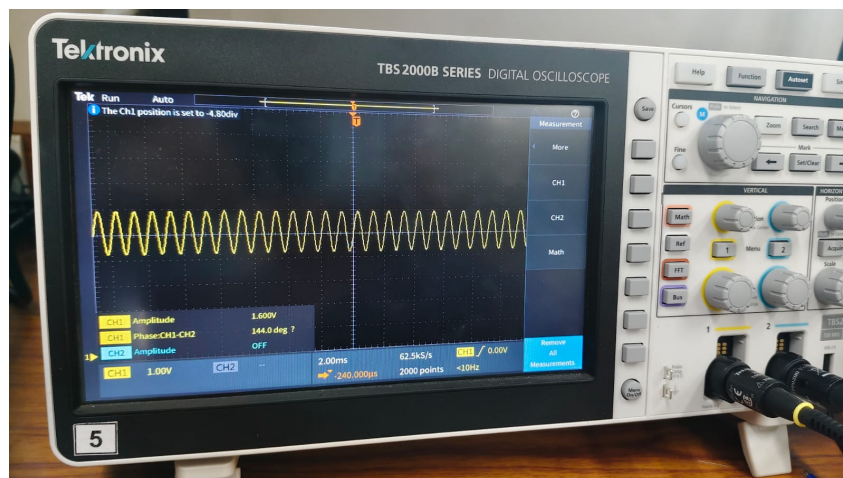


Figure 6: Experimental Plot

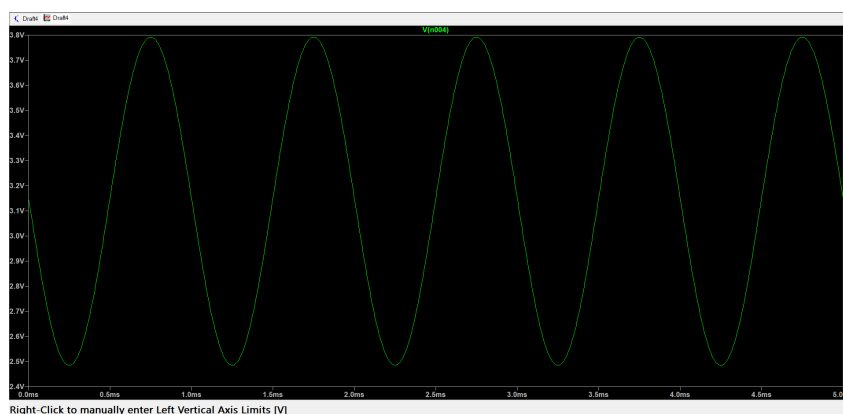


Figure 7: Simulation Plot

4.8.3 Deep Triode ($V_{in} = 2V$)

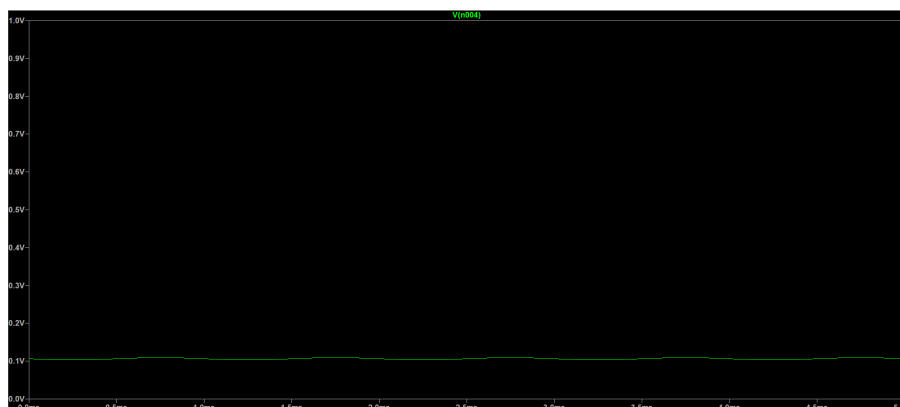


Figure 8: Simulation Plot

5 Analysis

5.1 Bias Point for Maximum Gain

The magnitude of the voltage gain is maximum at

$$\boxed{V_{IN,DC} = 1.72 \text{ V}}$$

where the measured output voltage is

$$V_{OUT,RMS} = 852 \text{ mV}.$$

Since the applied input signal is

$$V_{IN,RMS} = 10 \text{ mV},$$

the corresponding voltage gain is

$$A_v = -\frac{V_{OUT,RMS}}{V_{IN,RMS}} = -\frac{852}{10} = -85.2.$$

Hence,

$$\boxed{|A_v| = 85.2}.$$

The gain increases as the MOSFET is biased further above threshold because the transconductance increases with the operating current. The transconductance used for the analysis is

$$\boxed{g_m \approx 8.25 \text{ mS}}.$$

At the maximum-gain point, the MOSFET is operating in the high-gain region near the boundary between saturation and triode operation. As the input bias is increased beyond this point, the drain current increases and the drain voltage decreases. Therefore, the operating point moves closer to the saturation-to-triode boundary, causing the gain to decrease rapidly.

This behavior is clearly visible in the measured data. The gain falls from

$$|A_v| = 85.2 \quad \text{at } 1.72 \text{ V}$$

to

$$|A_v| = 54.4 \quad \text{at } 1.74 \text{ V}$$

and then to

$$|A_v| = 3.95 \quad \text{at } 1.76 \text{ V.}$$

Thus, the largest gain occurs at approximately 1.72 V, where the MOSFET has appreciable transconductance while still being close to the edge of saturation.

5.2 Gain in Cutoff and Deep Triode Regions

In the cutoff region, the MOSFET drain current is very small and the transconductance approaches zero:

$$g_m \rightarrow 0.$$

Since the small-signal gain is approximately

$$A_v \approx -g_m(R_D \parallel r_d),$$

the voltage gain also approaches zero. This explains the very small gain at low input bias. For example,

$$|A_v| = 0.669$$

at

$$V_{IN,DC} = 1.30 \text{ V.}$$

As the input bias is increased, the MOSFET begins conducting and g_m increases, resulting in a larger voltage gain.

At the other extreme, when the MOSFET enters deep triode operation, the drain voltage becomes low and the device no longer behaves as a good saturation-region current source. The effective small-signal gain therefore decreases strongly. In the measured data, the gain decreases very rapidly after 1.72 V:

$$|A_v| = 85.2 \quad \text{at } 1.72 \text{ V,}$$

$$|A_v| = 54.4 \quad \text{at } 1.74 \text{ V,}$$

and

$$|A_v| = 3.95 \quad \text{at } 1.76 \text{ V.}$$

Therefore, the gain approaches a small value both near cutoff and in deep triode operation, while the maximum occurs in the intermediate high-gain region.

5.3 Comparison of Method 1 and Method 2

Method 1 obtains the voltage gain from the slope of the measured DC transfer characteristic:

$$A_{v,1} \approx \frac{\Delta V_{OUT}}{\Delta V_{IN}}.$$

Method 2 obtains the gain directly from the small-signal AC response:

$$A_{v,2} = -\frac{V_{OUT,RMS}}{V_{IN,RMS}}.$$

At

$$V_{IN,DC} = 1.70 \text{ V},$$

Method 1 gives

$$A_{v,1} = -72.720$$

while Method 2 gives

$$A_{v,2} = -71.510.$$

The absolute difference is

$$|\Delta A_v| = |-72.720 - (-71.510)| = 1.210.$$

Therefore, the percentage difference is

$$\% \text{ difference} = \frac{1.210}{72.720} \times 100 \approx \boxed{1.66\%}.$$

Thus, the two methods agree very closely at 1.70 V.

At the maximum-gain bias point,

$$V_{IN,DC} = 1.72 \text{ V},$$

Method 2 gives

$$A_{v,2} = -85.2.$$

The available Method 1 value of -72.72 is the finite-difference slope over the 1.70 – 1.75 V interval, rather than an exact derivative at 1.72 V. Using this value for comparison,

$$|\Delta A_v| = |-85.2 - (-72.72)| = 12.48$$

and

$$\% \text{ difference} = \frac{12.48}{72.72} \times 100 \approx \boxed{17.16\%}.$$

The larger discrepancy near 1.72 V is expected because the transfer characteristic is highly nonlinear in this region. Method 1 uses a finite voltage interval and therefore gives an average slope, whereas Method 2 measures the local small-signal gain directly at the selected bias point. Additional differences can arise from measurement noise, bias-setting uncertainty, oscilloscope resolution, and the AC input amplitude not being sufficiently small compared with the nonlinear variation of the transfer characteristic.

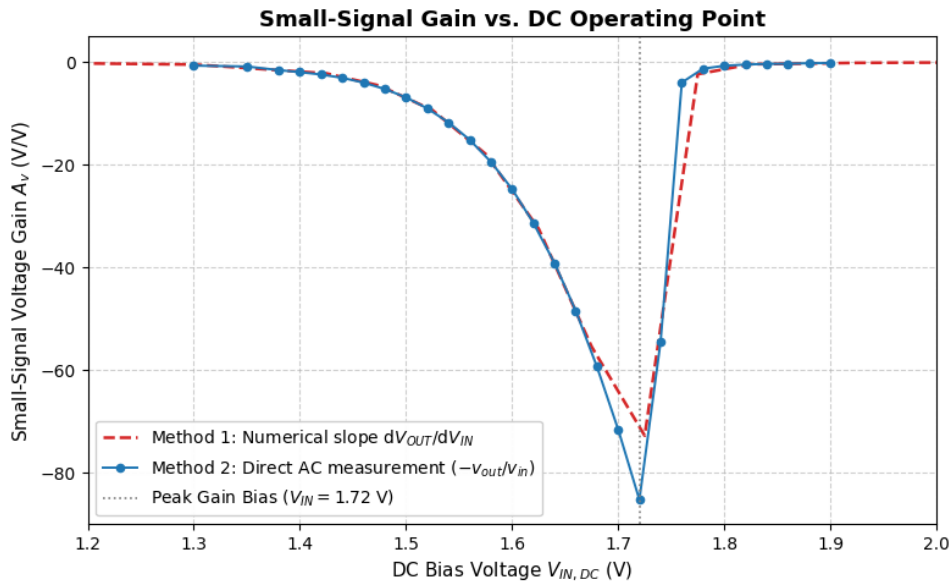


Figure 9: Comparison of Small-Signal Voltage Gain by Method 1 and Method 2

5.4 Effect of Increasing R_D

The small-signal voltage gain of the common-source amplifier can be approximated by

$$A_v \approx -g_m(R_D \parallel r_d).$$

Therefore, if R_D is increased while the MOSFET remains in saturation, the effective load resistance

$$R_D \parallel r_d$$

generally increases. Hence, the magnitude of the saturation-region voltage gain is expected to increase.

However, the drain voltage is given by

$$V_D = V_{DD} - I_D R_D.$$

For a larger R_D , a given drain current produces a larger voltage drop across the drain resistor. Consequently, the drain voltage falls more rapidly as the input bias is increased. The MOSFET therefore reaches the saturation-to-triode boundary at a lower drain current.

Thus, increasing R_D generally gives a higher saturation-region gain but reduces the available drain-voltage headroom and makes the practical saturation-region operating range narrower.

5.5 Relation Between the DC Transfer Slope and AC Gain

For a sufficiently small AC signal applied around a fixed DC operating point, the amplifier response can be linearized about that point. Therefore,

$$A_v = \frac{dv_{OUT}}{dv_{IN}}$$

which is exactly the local slope of the DC transfer characteristic. Hence, the slope of the large-signal DC transfer curve and the small-signal AC gain are the same at the same bias point.

5.6 Comparison with the Illustrative Section 6 Values

The illustrative simulation in Section 6 uses a different set of device parameters and is intended only to show the expected shape of the transfer and gain characteristics. From the illustrative plots, the threshold voltage is approximately

$$V_{TH, \text{illustrative}} \approx 2.0 \text{ V}.$$

For the measured MOSFET, the threshold voltage obtained using the onset method was

$$V_{TH, \text{measured}} = 1.40 \text{ V}.$$

Therefore, the difference is

$$\Delta V_{TH} = V_{TH, \text{measured}} - V_{TH, \text{illustrative}}$$

$$= 1.40 - 2.00 = -0.60 \text{ V}.$$

Thus, the measured device has a threshold voltage approximately

$$\boxed{0.60 \text{ V lower}}$$

than the illustrative device.

The illustrative gain curve reaches its maximum gain magnitude at an input DC bias of approximately

$$V_{IN,\text{peak,illustrative}} \approx 3.0 \text{ V}.$$

In the experimental measurements, the maximum gain occurs at

$$V_{IN,\text{peak,measured}} = 1.72 \text{ V},$$

where

$$|A_v| = 85.2.$$

Hence, the difference in the gain-peak bias voltage is

$$\Delta V_{IN,\text{peak}} = 1.72 - 3.0 = -1.28 \text{ V}.$$

Therefore, the maximum-gain point of the measured device occurs approximately

$$\boxed{1.28 \text{ V lower}}$$

than that of the illustrative device.

The measured device therefore has both a lower threshold voltage and a lower DC input voltage at which the maximum gain occurs. This difference is expected because the Section 6 simulation uses illustrative MOSFET parameters and is explicitly not intended as a numerical prediction of the measured device.

5.7 Comparison of Theoretical Gain with Part A and Part B

The theoretical small-signal voltage gain of the common-source amplifier was calculated using

$$A_v \approx -g_m(R_D \parallel r_d),$$

with

$$g_m = 8.25 \text{ mS}, \quad R_D = 10 \text{ k}\Omega, \quad r_d = 181.82 \text{ k}\Omega.$$

Therefore,

$$\boxed{|A_v|_{\text{theory}} \approx 78.2}.$$

In Part A, the gain was obtained from the slope of the DC transfer characteristic. The finite-difference slope near the maximum-gain region was

$$\boxed{|A_v|_{\text{Part A}} \approx 72.72}.$$

Taking the theoretical value as the reference, the percentage error is

$$\% \text{ Error}_{\text{Part A}} = \frac{|72.72 - 78.2|}{78.2} \times 100 \quad (8)$$

$$\approx \boxed{7.01\%}. \quad (9)$$

In Part B, the gain was measured directly using a small AC signal. At the maximum-gain bias point of 1.72 V, the measured gain was

$$\boxed{|A_v|_{\text{Part B}} = 85.2}.$$

The percentage error with respect to the theoretical value is

$$\% \text{ Error}_{\text{Part B}} = \frac{|85.2 - 78.2|}{78.2} \times 100 \quad (10)$$

$$\approx \boxed{8.95\%}. \quad (11)$$

Thus, the three gain values are

$$\boxed{|A_v|_{\text{Part A}} = 72.72, \quad |A_v|_{\text{Theory}} = 78.2, \quad |A_v|_{\text{Part B}} = 85.2}.$$

The theoretical value lies between the gains obtained from Part A and Part B:

$$72.72 < 78.2 < 85.2.$$

The error is approximately 7.01% for Part A and 8.95% for Part B. These differences are reasonable because the theoretical gain uses an approximate $g_m = 8.25 \text{ mS}$ and an r_d measured at a nearby bias point rather than exactly at 1.72 V. In addition, Part A uses a finite-difference slope of the nonlinear DC transfer curve, whereas Part B measures the local AC response and may be affected by measurement noise, bias-point uncertainty,

and finite AC signal amplitude.

6 Sources of Error

The following factors may contribute to differences between the measured and theoretical results:

1. The DC bias voltage may not have been set exactly at the desired value.
2. The AC input signal has a finite amplitude, so the small-signal approximation may not be exact.
3. Oscilloscope noise and measurement resolution can introduce errors, especially when the output voltage is small.
4. Method 1 uses finite voltage intervals to estimate the slope, which can differ from the local derivative when the transfer characteristic changes rapidly.

7 Conclusion

The small-signal voltage gain of the NMOS common-source amplifier was successfully measured using the direct AC measurement method.

The gain was found to depend strongly on the DC operating point. At lower input voltages, the transistor is close to cutoff and the gain is small. As the input voltage increases, the transconductance increases and the voltage gain increases rapidly.

The maximum measured gain was obtained at

$$V_{IN,DC} = 1.72 \text{ V}$$

with

$$V_{OUT,RMS} = 852 \text{ mV}$$

and

$$A_v = -85.2.$$

After the maximum gain point, the gain decreases rapidly as the transistor moves towards and into the triode region.

The experiment also demonstrates the relationship between the DC transfer characteristic and small-signal AC behavior. For a sufficiently small signal, the AC gain is equal to the local slope of the DC transfer characteristic:

$$A_v = \left. \frac{dV_{OUT}}{dV_{IN}} \right|_Q. \quad (12)$$

Therefore, the DC transfer characteristic can be used to understand and estimate the small-signal behavior of the NMOS common-source amplifier.